Connexive Logics | 8

8.1 Philosophical Motivations

Connexive logics are motivated by some dissatisfaction with standard accounts of implication. A central intuition is that a proposition should not imply its own negation, nor should its negation imply the proposition itself.

In other words, we want that inferences of the following form be logically valid:

$$/ \neg(\neg \mathtt{A} \rightarrow \mathtt{A}) \tag{AT1}$$

Connexive logics are logical systems in which this and other similar inference forms are logically valid.

A second important motivation is historical. Connexive theses are commonly associated with the theses about the conditional by several authors such as Aristotle, Chrysippus, and Boethius. Many researchers regard connexive logic as a way of recovering and systematising ideas that can be traced back to ancient and medieval traditions of logical thought.

More generally, connexive logics are motivated by the desire to provide a more demanding notion of implication. According to this perspective, a genuine implication should involve some form of coherence or compatibility between antecedent and consequent.

8.2 Vocabularies and Languages

The vocabulary and formal language $\mathbf{Fo}_{\mathsf{W0}}$ and $\mathbf{Fo}_{\mathsf{W0}}$ will be defined as those of $\mathsf{S0}$.

Definition 8.1.

$$\mathbf{Vo}_{\mathsf{W0}} \overset{\text{DF}}{=} \mathbf{Vo}_{\mathsf{S0}}. \qquad \mathbf{Fo}_{\mathsf{W0}} \overset{\text{DF}}{=} \mathbf{Fo}_{\mathsf{S0}}.$$

8.3 Inference and Consequence Relations

$\mathsf{W0}$ falls among modern systems, so their inference relations are just special cases of definition 3.31.

As for the contents of their consequence relations, we want that all inferences of these forms be valid:

$$/ \neg(\neg A \rightarrow A) \tag{AT1}$$

$$/ \neg(A \rightarrow \neg A) \tag{AT2}$$

$$(A \rightarrow B) / \neg(A \rightarrow \neg B) \tag{BT1}$$

$$(A \rightarrow \neg B) / \neg(A \rightarrow B) \tag{BT2}$$

Since none of these inference forms are valid in S0, the classical logic, we say that W0 is a *contra-classical logic*. A contra-classical logic is a non-classical logic which not reject or restrict some of its theses, but also propose theses which are not part of classical logic.

Moreover, following McCall (1966, p. 416), we do not want all inferences of the following form to be valid in a connexive logic:

$$/ ((A \rightarrow B) \rightarrow (B \rightarrow A)).$$

This inference form is already non-valid in S0, se we must only be careful not to define W0 so that it becomes valid.

For the rest, we want W0 to be very much like S0, as this system is, after all, inspired by many of the intuitions on which classical logic is also inspired.

8.4 Interpretation Systems

The semantics of this system involves possible worlds, and we will not develop it in full.

It is important to remark, however, that this semantics treats truth and falsity conditions independently.

8.5 Derivation Systems

The derivation system originally proposed for W0 is an axiomatic one, which is also based on the positive logic. Such system is defined by the following inference forms:

$$(\mathtt{A} \longrightarrow \mathtt{B}), \mathtt{A} \,/\, \mathtt{B} \qquad \text{(MPP)}$$
$$/\, (\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{A})) \qquad \text{(P1)}$$
$$/\, ((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C})) \longrightarrow (\mathtt{A} \longrightarrow \mathtt{C}))) \qquad \text{(P2)}$$
$$/\, (\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow (\mathtt{A} \wedge \mathtt{B}))) \qquad \text{(P3)}$$
$$/\, ((\mathtt{A} \wedge \mathtt{B}) \longrightarrow \mathtt{A}) \qquad \text{(P4)}$$
$$/\, ((\mathtt{A} \wedge \mathtt{B}) \longrightarrow \mathtt{B}) \qquad \text{(P5)}$$
$$/\, (\mathtt{A} \longrightarrow (\mathtt{A} \vee \mathtt{B})) \qquad \text{(P6)}$$
$$/\, (\mathtt{B} \longrightarrow (\mathtt{A} \vee \mathtt{B})) \qquad \text{(P7)}$$
$$/\, ((\mathtt{A} \longrightarrow \mathtt{C}) \longrightarrow ((\mathtt{B} \longrightarrow \mathtt{C}) \longrightarrow ((\mathtt{A} \vee \mathtt{B}) \longrightarrow \mathtt{C}))) \qquad \text{(P8)}$$
$$/\, (\neg\neg\mathtt{A} \leftrightarrow \mathtt{A}) \qquad \text{(DN)}$$
$$/\, (\neg(\mathtt{A} \vee \mathtt{B}) \leftrightarrow (\neg\mathtt{A} \wedge \neg\mathtt{B})) \qquad \text{(DM1)}$$
$$/\, (\neg(\mathtt{A} \wedge \mathtt{B}) \leftrightarrow (\neg\mathtt{A} \vee \neg\mathtt{B})) \qquad \text{(DM2)}$$
$$/\, (\neg(\mathtt{A} \longrightarrow \mathtt{B}) \leftrightarrow (\mathtt{A} \longrightarrow \neg\mathtt{B})) \qquad \text{(W0)}$$

As usual, $\ulcorner(\mathtt{A} \leftrightarrow \mathtt{B})\urcorner$ is an abbreviation of $\ulcorner((\mathtt{A} \longrightarrow \mathtt{B}) \wedge (\mathtt{B} \longrightarrow \mathtt{A}))\urcorner$ in the context of these axioms.

Note that the axioms added to positive logic in this system are satisfied by S0, except for (W0), which W0's characteristic axiom. Thus, an axiomatic deductive base for W0 would consist of P0's plus (DN), (DM1), (DM2), and (W0).

8.5.1 Fitch-style derivations for W0

To present the deductive base $\mathbf{Cal}_{\mathsf{W0}}$ for the connexive propositional logic W0 in the style of notation 3.48, we will, again, extend that of $\mathbf{Cal}_{\mathsf{P0}}$. First, we will add rules corresponding to De Morgan rules and double negation:

Double negation rules.

$$\begin{array}{l|ll} a & \mathtt{A} & \\ & \vdots & \\ & \neg\neg\mathtt{A} & \neg\neg^{+}, a \end{array} \qquad\qquad \begin{array}{l|ll} a & \neg\neg\mathtt{A} & \\ & \vdots & \\ & \mathtt{A} & \neg\neg^{-}, a \end{array}$$

De Morgan rules.

$$\begin{array}{r|ll} a & \neg(\mathtt{A} \vee \mathtt{B}) & \\ & \vdots & \\ & (\neg\mathtt{A} \wedge \neg\mathtt{B}) & \mathsf{DM}_1^-, a \end{array} \qquad \begin{array}{r|ll} a & (\neg\mathtt{A} \wedge \neg\mathtt{B}) & \\ & \vdots & \\ & \neg(\mathtt{A} \vee \mathtt{B}) & \mathsf{DM}_1^+, a \end{array}$$

$$\begin{array}{r|ll} a & \neg(\mathtt{A} \wedge \mathtt{B}) & \\ & \vdots & \\ & (\neg\mathtt{A} \vee \neg\mathtt{B}) & \mathsf{DM}_2^-, a \end{array} \qquad \begin{array}{r|ll} a & (\neg\mathtt{A} \vee \neg\mathtt{B}) & \\ & \vdots & \\ & \neg(\mathtt{A} \wedge \mathtt{B}) & \mathsf{DM}_2^+, a \end{array}$$

These rules easily allow us to derive axioms (DN), (DM1), and (DM2). As for axiom (W0), we need to add a the following rules:

Connexive rules.

$$\begin{array}{r|ll} \begin{array}{r} a \\ \\ b \end{array} & \begin{array}{|l} \underline{\mathtt{A}} \\ \vdots \\ \neg\mathtt{B} \end{array} & \\ & \vdots & \\ & \neg(\mathtt{A} \longrightarrow \mathtt{B}) & \not\twoheadrightarrow^+, a\text{–}b \end{array} \qquad \begin{array}{r|ll} a & \neg(\mathtt{A} \longrightarrow \mathtt{B}) & \\ & \vdots & \\ b & \mathtt{A} & \\ & \vdots & \\ & \neg\mathtt{B} & \not\twoheadrightarrow^-, a, b \end{array}$$

Thus, we might define $\mathbf{DBas}_{\mathsf{W0}}$ as follows:

Definition 8.2 (Deductive base for W0)**.**

$$\mathbf{DBas}_{\mathsf{W0}} \stackrel{\text{DF}}{=} \mathbf{DBas}_{\mathsf{P0}} \cup \{\neg\neg^-, \neg\neg^+, \mathsf{DM}_1^-, \mathsf{DM}_2^-, \mathsf{DM}_1^+, \mathsf{DM}_2^+, \not\twoheadrightarrow^-, \not\twoheadrightarrow^+\}.$$

Note that no specific rules for negation or the falsum were added.

As for the derivation system $\mathbf{Cal}_{\mathsf{W0}}$, we will once again assume that the functions $\mathbf{Prem}_{\mathsf{W0}}$ and $\mathbf{Conc}_{\mathsf{W0}}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 8.3 ($\mathbf{Cal}_{\mathsf{W0}}$)**.**

$\mathbf{Cal}_{\mathsf{W0}}$ is a calculus on $\mathsf{W0}$

$$\overset{\text{DF}}{=} \mathbf{Cal}_{\mathsf{W0}} = \langle \mathbf{DBas}_{\mathsf{W0}}, \mathbf{Der}_{\mathsf{W0}}, \mathbf{Prem}_{\mathsf{W0}}, \mathbf{Conc}_{\mathsf{W0}}, \vdash_{\mathsf{W0}} \rangle$$
$$: \mathbf{DBas}_{\mathsf{W0}} \text{ is a deductive base on } \mathbf{Fo}_{\mathsf{W0}} \mathbin{\wedge\!\!\!\wedge}$$
$$\mathbf{Der}_{\mathsf{W0}} = \{\mathbf{D} : \mathbf{D} \text{ is a Fitch derivation on } \mathbf{Fo}_{\mathsf{W0}} \text{ from } \mathbf{DBas}_{\mathsf{W0}}\} \mathbin{\wedge\!\!\!\wedge}$$
$$\mathbf{Prem}_{\mathsf{W0}} : \mathbf{Der}_{\mathsf{W0}} \longrightarrow \wp\mathbf{Fo}_{\mathsf{W0}} \mathbin{\wedge\!\!\!\wedge}$$
$$\mathbf{Conc}_{\mathsf{W0}} : \mathbf{Der}_{\mathsf{W0}} \longrightarrow \wp\mathbf{Fo}_{\mathsf{W0}} \mathbin{\wedge\!\!\!\wedge}$$
$$\vdash_{\mathsf{S}} = \{\langle \mathbf{A}, \mathbf{B} \rangle : \exists \mathbf{X} \in \mathbf{Der}_{\mathsf{W0}} \, (\mathbf{Prem}_{\mathsf{W0}}(\mathbf{X}) = \mathbf{A} \mathbin{\wedge\!\!\!\wedge} \mathbf{Conc}_{\mathsf{W0}}(\mathbf{X}) = \mathbf{B})\}.$$

Task 8.1. Show that all inferences of the forms (DN), (DM1), (DM2), and (W0) are members of $\vdash_{\mathsf{W0}}$.

The cases (DN), (DM1), and (DM2) are easy. Let us, then, prove the case of the axiom schema distinctive of W0, i.e., (W0).

Theorem 8.4. $\vdash_{\mathsf{W0}} (\neg(\mathtt{A} \longrightarrow \mathtt{B}) \leftrightarrow (\mathtt{A} \longrightarrow \neg\mathtt{B}))$.

Proof. Let $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{\mathsf{W0}}$ and consider the schematic derivation:

1	$\mid \quad \mid\ \neg(\mathtt{A} \longrightarrow \mathtt{B})$	
2	$\mid \quad \mid \quad \mid\ \mathtt{A}$	
3	$\mid \quad \mid \quad \mid\ \neg\mathtt{B}$	$\not\rightarrow^{-}$, 1, 2
4	$\mid \quad \mid\ (\mathtt{A} \longrightarrow \neg\mathtt{B})$	$\longrightarrow^{+}$, 2–3
5	$\mid\ (\neg(\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow (\mathtt{A} \longrightarrow \neg\mathtt{B}))$	$\longrightarrow^{+}$, 2–4
6	$\mid \quad \mid\ (\mathtt{A} \longrightarrow \neg\mathtt{B})$	
7	$\mid \quad \mid \quad \mid\ \mathtt{A}$	
8	$\mid \quad \mid \quad \mid\ \neg\mathtt{B}$	$\longrightarrow^{-}$, 6, 7
9	$\mid \quad \mid\ \neg(\mathtt{A} \longrightarrow \mathtt{B})$	$\not\rightarrow^{+}$, 7–8
10	$\mid\ ((\mathtt{A} \longrightarrow \neg\mathtt{B}) \longrightarrow \neg(\mathtt{A} \longrightarrow \mathtt{B}))$	$\longrightarrow^{+}$, 6–9
11	$\mid\ (\neg(\mathtt{A} \longrightarrow \mathtt{B}) \leftrightarrow (\mathtt{A} \longrightarrow \neg\mathtt{B}))$	$\wedge^{+}$, 5, 10

Call this schematic derivation '**D**'. It is clear, for all $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{\mathsf{W0}}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{\mathsf{W0}}$ from $\mathbf{DBas}_{\mathsf{W0}}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\mathsf{W0}}$ with $\{\} \subseteq \mathbf{Prem}_{\mathsf{W0}}(\mathbf{D})$ and $\ulcorner(\neg(\mathtt{A} \longrightarrow \mathtt{B}) \leftrightarrow (\mathtt{A} \longrightarrow \neg\mathtt{B}))\urcorner \in \mathbf{Conc}_{\mathsf{W0}}(\mathbf{D})$. Therefore, by definition 8.3, $\vdash_{\mathsf{W0}} (\neg(\mathtt{A} \longrightarrow \mathtt{B}) \leftrightarrow (\mathtt{A} \longrightarrow \neg\mathtt{B}))$. $\square$

Similarly, all instances of (AT1), (AT2), (BT1), and (BT2) are derivable in this $\mathbf{Der}_{\mathsf{W0}}$.

Task 8.2. Show that all inferences of the forms (AT1), (AT2), (BT1), and (BT2) are members of $\vdash_{W0}$.

We will prove the cases of (AT1) and (BT1).

Theorem 8.5. $\vdash_{W0} \neg(\neg A \longrightarrow A)$

Proof. Let $A \in \mathbf{Fo}_{W0}$ and consider the schematic derivation:

$$
\begin{array}{r|l|ll}
1 & & \neg A & \\
\cline{3-3}
2 & & \neg A & \text{Reit, } 1 \\
3 & \multicolumn{2}{l}{\neg(\neg A \longrightarrow A)} & \not\rightarrow^{+}, 1\text{–}2
\end{array}
$$

Call this schematic derivation '**D**'. It is clear, for all $A \in \mathbf{Fo}_{W0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{W0}$ from $\mathbf{DBas}_{W0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{W0}$ with $\{\} \subseteq \mathbf{Prem}_{W0}(\mathbf{D})$ and $\ulcorner\neg(\neg A \longrightarrow A)\urcorner \in \mathbf{Conc}_{W0}(\mathbf{D})$. Therefore, by definition 8.3, $\vdash_{W0} \neg(\neg A \longrightarrow A)$. □

Theorem 8.6. $(A \longrightarrow B) \vdash_{W0} \neg(A \longrightarrow \neg B)$

Proof. Let $A, B \in \mathbf{Fo}_{W0}$ and consider the schematic derivation:

$$
\begin{array}{r|l|ll}
1 & \multicolumn{2}{l}{(A \longrightarrow B)} & \\
\cline{2-3}
2 & & A & \\
\cline{3-3}
3 & & B & \longrightarrow^{-}, 1, 2 \\
4 & & \neg\neg B & \neg\neg^{+}, 3 \\
5 & \multicolumn{2}{l}{\neg(A \longrightarrow \neg B)} & \not\rightarrow^{+}, 2\text{–}4
\end{array}
$$

Call this schematic derivation '**D**'. It is clear, for all $A, B \in \mathbf{Fo}_{W0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{W0}$ from $\mathbf{DBas}_{W0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{W0}$ with $\{\ulcorner(A \longrightarrow B)\urcorner\} \subseteq \mathbf{Prem}_{W0}(\mathbf{D})$ and $\ulcorner\neg(A \longrightarrow \neg B)\urcorner \in \mathbf{Conc}_{W0}(\mathbf{D})$. Therefore, by definition 8.3, we have that $(A \longrightarrow \neg B) \vdash_{W0} \neg(A \longrightarrow B)$. □

A curious fact about W0 is that it is a *contradictory logic*, i.e., it is a logic which has at least one formula A such that both $\mathtt{A}$ and $\ulcorner\neg\mathtt{A}\urcorner$ are theorems of W0.

Task 8.3. Show that $\vdash_{\mathrm{W0}}$ has contradictory theorems, i.e., show that, for some $\mathtt{A} \in \mathbf{Fo}_{\mathrm{W0}}$, the inferences $/\ \mathtt{A}$ and $/\ \neg\mathtt{A}$ are members of $\vdash_{\mathrm{W0}}$.

Theorem 8.7. $\vdash_{\mathrm{W0}}$ has contradictory theorems.

Proof. Let $\mathtt{A} \in \mathbf{Fo}_{\mathrm{W0}}$. We show that the formula $\ulcorner(\mathtt{A} \wedge \neg\mathtt{A})\urcorner$ and its negation $\ulcorner\neg(\mathtt{A} \wedge \neg\mathtt{A})\urcorner$ are derivable in $\vdash_{\mathrm{W0}}$.

For the first, consider the schematic derivation:

1	$(\mathtt{A} \wedge \neg\mathtt{A})$	
2	$\mathtt{A}$	$\wedge_1^-$, 1
3	$((\mathtt{A} \wedge \neg\mathtt{A}) \rightarrow \mathtt{A})$	$\rightarrow^+$, 1–2

Call this schematic derivation '**D1**'. It is clear, for all $\mathtt{A} \in \mathbf{Fo}_{\mathrm{W0}}$, that **D1** is a Fitch derivation on $\mathbf{Fo}_{\mathrm{W0}}$ from $\mathbf{DBas}_{\mathrm{W0}}$. This means that there is a derivation $\mathbf{D1} \in \mathbf{Der}_{\mathrm{W0}}$ with $\{\} \subseteq \mathbf{Prem}_{\mathrm{W0}}(\mathbf{D1})$ and $\ulcorner((\mathtt{A} \wedge \neg\mathtt{A}) \rightarrow \mathtt{A})\urcorner \in \mathbf{Conc}_{\mathrm{W0}}(\mathbf{D1})$. Hence, by definition 8.3, we have that $\vdash_{\mathrm{W0}} ((\mathtt{A} \wedge \neg\mathtt{A}) \rightarrow \mathtt{A})$.

For the contradictory, consider this other schematic derivation:

1	$(\mathtt{A} \wedge \neg\mathtt{A})$	
2	$\neg\mathtt{A}$	$\wedge_2^-$, 1
3	$\neg((\mathtt{A} \wedge \neg\mathtt{A}) \rightarrow \mathtt{A})$	$\nrightarrow^+$, 1–2

Call this schematic derivation '**D2**'. It is clear, for all $\mathtt{A} \in \mathbf{Fo}_{\mathrm{W0}}$, that **D2** is a Fitch derivation on $\mathbf{Fo}_{\mathrm{W0}}$ from $\mathbf{DBas}_{\mathrm{W0}}$. This means that there is a derivation $\mathbf{D2} \in \mathbf{Der}_{\mathrm{W0}}$ with $\{\} \subseteq \mathbf{Prem}_{\mathrm{W0}}(\mathbf{D2})$ and $\ulcorner((\mathtt{A} \wedge \neg\mathtt{A}) \rightarrow \mathtt{A})\urcorner \in \mathbf{Conc}_{\mathrm{W0}}(\mathbf{D2})$. Hence, by definition 8.3, we have that $\vdash_{\mathrm{W0}} \neg((\mathtt{A} \wedge \neg\mathtt{A}) \rightarrow \mathtt{A})$.

Note that $\ulcorner((\mathtt{A} \wedge \neg\mathtt{A}) \rightarrow \mathtt{A})\urcorner$ and $\ulcorner\neg((\mathtt{A} \wedge \neg\mathtt{A}) \rightarrow \mathtt{A})\urcorner$, both members of W0, are mutually contradictory. Therefore, $\vdash_{\mathrm{W0}}$ has contradictory theorems. □

Despite this inconsistency, the system is non-trivial.